

Total No. of Questions : 4]

SEAT No. :

PD3

[Total No. of Pages : 2

[6408]-103

F.E. (Common) (Insem)

ENGINEERING PHYSICS

(2019 Pattern) (Semester-II) (Credit System) (107002)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Solve (Q.1 OR Q.2) AND (Q.3 OR Q.4)
- 2) Figures to the right indicate full marks.
- 3) Use of logarithmic tables, slide rule, electronic pocket calculator is allowed.
- 4) Assume suitable data, if necessary.

- Q1)** a) By using neat ray diagram, derive the expression of amplitude for Fraunhofer's diffraction due to single slit. [6]
- b) Explain the use of thin film interference to design anti reflection coating. [5]
- c) At what angle should the polarizer and analyzer be oriented so that intensity of natural beam of light is (i) 20% and (ii) 60% of its original intensity. [4]

OR

- Q2)** a) When uniform thin film is illuminated with monochromatic light, derive expression for the effective path difference for reflected system. [6]
- b) State and prove law of Malus. [5]
- c) Find the angular position of first minimum in the Fraunhofer diffraction pattern due to single slit of width $6.6 \times 10^{-7}\text{m}$, when illuminated by light of wavelength 6000 Å. [4]

- Q3)** a) Explain Construction and working of CO_2 laser. [6]
- b) What is optical fiber? Differentiate between step index & graded index optical fibers. (any 4 points). [5]
- c) Define the following and only write down its equations- [4]
- i) Acceptance angle
 - ii) Numerical aperture for optical fiber.

OR

P.T.O.

Q4) a) Define: [6]

- i) Critical angle
- ii) Total internal reflection

In an optical fiber, the core material has R.I. 1.49 and R.I. of clad material is 1.44.

Calculate the value of maximum entrance angle & Numerical aperture.

- b) Explain the use of laser for hologram recording and reconstruction. [5]
- c) What is stimulated emission of radiations? Explain its role in the laser system. [4]

